Higher National Diploma in Engineering (Electrical)

4th Year, First Semester Examination – 2019

EE 3214 – Construction and Protection of Electrical systems

Instructions for Candidates:	No. of questions : 06
answer **only five** questions.	No. of pages : 03
each question carry equal marks	Time : **03 hours**

1.

I. What is the advantage of using per unit calculations for the solution of transformer problems? (2 Marks)

II. An Electrical heater is rated 200V, 500 Watts. If the voltage applied across the lamp is twice the rated value, calculate the current that flows though the heater **using per unit method**. (5 Marks)

III. A 3 phase generating station has two generators rated 10,000 kVA and 5,000 kVA with 12% and 18% reactance respectively. Generators are connected a 12,000V bus bar from which the load is taken through a 2000kVA step up transformer having a reactance of 6%. If a fault occur at the high voltage end of the transformer, what is the short circuit MVA (Fault MVA) at that point? (13 Marks)

2.

I. Define the operator "a "which are related with unsymmetrical fault calculation and find a value for $(a^2 + a + 1)$.

(4 Marks)

II. A balanced star connected load takes 240A from a balanced 3 phase 4 wire supply. If fuses in the B and Y phases are blown find the symmetrical components of the line currents in all 3 phases.

(16 Marks)

3.

I. What is known as 'sensitivity' and 'selectivity' of a protective relay? (4 Marks)

II. What is the operating principle of a differential relay? (3 Marks)

III. Define current setting of a relay (2 Marks)

IV. What is the use of the 'Time setting multiplier' in a relay (2 Marks)

V. An over current relay is connected to a 250/30 current transformer, having a current setting of 150% that has primary fault current of 860A. The time setting multiplier is 0.8. Time/PSM curve is given on figure Q3.1. Determine, (9 Marks)

a) The Plug setting multiplier

b) The actual relay operating time.

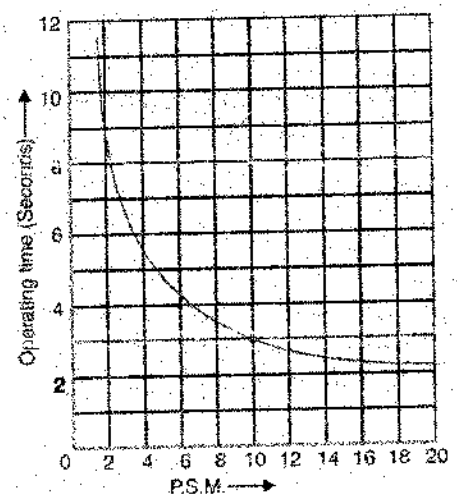

Figure Q3.1

4.

I. What is known as an 'Arcing ground'? (2 Marks)

II. What are the most commonly used methods to provide protection against lighting surges? (3 Marks)

III. Explain the operation of a transformer Earth- Fault or Leakage protection relay using suitable diagram. (7 Marks)

IV. A 3 phase transformer with 230/6600 line voltage is connected in star/delta configuration. The protection transformer on 230V side have a current ratio of 30:1.

a. Draw the Merz-Price circulation protection scheme for above fault

b. What must be the current transformer ratio on 6600 V side? (8 Marks)

5.

I. Explain the 'Fault bus protection' scheme using a suitable diagram. (8 Marks)

II. Write three common methods used for line protection (3 Marks)

III. What ways can the time graded protection be achieved in a parallel feeder? (5 Marks)

IV. Explain the principle and operation of distance relay protection which is used for transmission line protection (4 Marks)

6.

I. What kind of protection scheme is used when an 'over speed' situation occur in an alternator? (4 Marks)

II. Explain the protection scheme used for 'unbalanced loading' of three phase alternator using a diagram. (5 Marks)

III. A star connected 3 phase 15MVA, 11kV alternator has a per phase reactance of 10%. It is protected by Merz price circulation current principle which is set to operate for fault currents not less than 200A. Calculate the value of earthing resistance to be provided in order to ensure that only 10% of the alternator winding remains unprotected. (11 Marks)